Gas exchange in humans

Mark scheme and worked solutions for 20 questions: 10 multiple choice, 10 structured.

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1 · Part 1

Core | Gas exchange in humans | [1]

Answer: B

Explanation

Gas exchange surfaces in humans have a large surface area, a thin surface, a good blood supply and good ventilation with air. Option A has a thick surface, option C has a small surface area and option D has a poor blood supply, so each contains one wrong feature.

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2 · Part 1

Core | Gas exchange in humans | [1]

Answer: D

Explanation

Fresh blood arriving at the alveoli has a low oxygen concentration, so a concentration gradient between the air in the alveoli and the blood is kept, and oxygen keeps diffusing into the blood. Options A and B describe other features (thin surface, large surface area) that are not caused by blood flow. Option C is wrong because oxygen moves by diffusion, not by being pumped.

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3 · Part 1

Core | Gas exchange in humans | [1]

Answer: A

Explanation

Expired air contains less oxygen, more carbon dioxide and more water vapour than inspired air. Option B wrongly gives less water vapour. Options C and D reverse the oxygen and carbon dioxide changes.

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4 · Part 1

Core | Gas exchange in humans | [1]

Answer: C

Explanation

Limewater turns milky with carbon dioxide. Expired air has more carbon dioxide than inspired air, so the limewater in tube 1 turns milky first. Option A reverses the result. Option B is wrong because breathing does change the carbon dioxide content of air. Option D gives the right observation but a wrong conclusion: limewater tests for carbon dioxide, not oxygen.

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5 · Part 1 Core Gas exchange in humans [1]

Answer: A

Explanation

The breathing rate rose from 14 to 30 breaths per minute and the volume per breath (depth) rose from 0.5 dm^3 to 1.8 dm^3, so both increased. Options B, C and D each describe one of the two changes wrongly.

6 · Part 1 Extended Gas exchange in humans [1]

Answer: D

Explanation

The rings of cartilage support the trachea and keep it open so that air can flow freely, even when the pressure inside it falls during breathing in. Trapping dust and pathogens (A) is the job of mucus, moving mucus (B) is the job of cilia, and gas exchange (C) happens in the alveoli, not the trachea.

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Answer: B

Explanation

When breathing in, the external intercostal muscles contract and the diaphragm contracts and flattens, so the volume of the thorax increases and the pressure inside it decreases below atmospheric pressure. Option A describes breathing out. Option C wrongly says the diaphragm relaxes and the pressure increases. Option D wrongly says the external intercostal muscles relax and the volume decreases.

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Answer: C

Explanation

The alveoli are lined with a thin film of moisture; water evaporates from it into the air in the alveoli, so expired air contains more water vapour. Option A gives the wrong direction: oxygen diffuses from the air into the blood. Option B is wrong because oxygen is not turned into carbon dioxide in the alveoli; oxygen is absorbed into the blood and used in respiration in cells. Option D gives the wrong direction: carbon dioxide diffuses from the blood into the air in the alveoli.

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9 · Part 1 Extended Gas exchange in humans [1]

Answer: B

Explanation

During physical activity muscles respire faster, so more carbon dioxide passes into the blood. The brain detects this increased carbon dioxide concentration and causes an increased rate and greater depth of breathing. Option A has the change the wrong way round. Option C is wrong because the lungs are not the detector and oxygen in the alveoli does not increase. Option D is wrong because the brain, not the muscles, detects the change.

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Answer: A

Explanation

Goblet cells make mucus, which traps pathogens and particles; ciliated cells have cilia that beat to move the mucus up towards the throat. Option B swaps the two roles. Options C and D describe phagocytosis, which is done by white blood cells, not goblet cells or ciliated cells; option D also has the mucus moving the wrong way.

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11 · Part 1 Core Gas exchange in humans [5]

Mark scheme

(a) **[2]**

- thin (surface) / short diffusion distance (1)
- good blood supply / many capillaries (1)
- good ventilation with air (1)
- max 2

ALLOW moist surface

REJECT large surface area (excluded by the question)

(b) **[1]**

- more oxygen / carbon dioxide can diffuse at the same time / faster rate of diffusion (1)

(c) **[2]**

- surface area (for gas exchange) is reduced (1)
- so less oxygen diffuses into the blood / less oxygen reaches the muscles for (aerobic) respiration (1)

ALLOW less carbon dioxide removed from the blood

Worked solution

(a) Human gas exchange surfaces have four features: a large surface area, a thin surface, a good blood supply and good ventilation with air. The question excludes the large surface area, so any two of the other three score. (b) Gases cross the surface by diffusion. The bigger the area, the more molecules can diffuse across at the same time, so gas exchange is faster. (c) Fewer, larger air spaces have a smaller total surface area than many small alveoli. Less oxygen can diffuse into the blood each minute, so the muscles get less oxygen for respiration when walking and the person becomes out of breath.

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12 · Part 1 Core Gas exchange in humans [6]

Mark scheme

(a) **[2]**

- expired air contains less oxygen (1)
- expired air contains more carbon dioxide (1)
- maximum 1 mark if no data from the table are quoted (e.g. oxygen 21% to 16%, carbon dioxide 0.04% to 4%)

(b) **[2]**

- 4 ÷ 0.04 (1)
- 100 (times) (1)

(c) **[1]**

- expired air contains more water vapour (1)

(d) **[1]**

- expired air contains a lot of water vapour, which condenses (into droplets) in the cold air (1)

ALLOW water vapour in breath cools and turns into liquid water

Worked solution

(a) Compare each row of the table: oxygen falls from 21% in inspired air to 16% in expired air, and carbon dioxide rises from 0.04% to 4%. A full answer gives both changes with figures. (b) Divide the expired value by the inspired value: 4 ÷ 0.04 = 100, so there is 100 times as much carbon dioxide in expired air. Check: 0.04 × 100 = 4. (c) Expired air contains more water vapour than inspired air. (d) Because breath contains a lot of water vapour, when it meets cold air the vapour cools and condenses into tiny droplets, which can be seen as a cloud.

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13 · Part 1 Core Gas exchange in humans [6]

Mark scheme

(a) **[1]**

- carbon dioxide (1)

(b) **[2]**

- expired air contains more carbon dioxide than inspired air (1)
- inspired air contains very little carbon dioxide / not enough to turn the limewater milky in 40 breaths (1)

> **ALLOW** tube Q turned milky after only 6 breaths, tube P had not turned milky after 40 breaths (for 1 mark if no conclusion is given)

(c) **[1]**

- concentration of the limewater / size or diameter of the tubes / depth of the tubing below the surface of the limewater / temperature (1)

> **ALLOW** volume of air per breath / depth of breathing

> **REJECT** volume of limewater (excluded by the question); number of breaths

(d) **[1]**

- limewater only tests for carbon dioxide / limewater does not react with oxygen (1)

(e) **[1]**

- inspired air contains a small amount of carbon dioxide (which builds up in the limewater over time) (1)

> **ALLOW** her expired air increases the carbon dioxide concentration of the air in the room (which is then drawn through tube P)

Worked solution

(a) Limewater turns milky (cloudy) when carbon dioxide passes through it. (b) Tube Q received expired air and turned milky after only 6 breaths; tube P received air from the room (inspired air) and was still clear after 40 breaths. So expired air contains much more carbon dioxide than inspired air, which contains very little. (c) For a fair test, everything except the type of air should be the same, for example the concentration of the limewater or the size of the tubes. (d) Limewater changes only when carbon dioxide is present, so it gives no information about oxygen. (e) Inspired air is not free of carbon dioxide; it contains about 0.04%. If enough of it is bubbled through, the carbon dioxide eventually turns the limewater milky.

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14 · Part 1 Core Gas exchange in humans [4]

Mark scheme

(a) **[1]**

- 38 (breaths per minute) (1)

(b) **[2]**

- breathing rate increases, from 16 to 38 breaths per minute / from 8 to 19 breaths in 30 s (1)
- depth of breathing increases / breaths become deeper (1)

(c) **[1]**

- repeat the investigation (several times) and calculate a mean (1)

ALLOW repeat with more people and calculate a mean

Worked solution

(a) The breaths were counted for 30 seconds, which is half a minute, so double the count: 19 × 2 = 38 breaths per minute. Check: 38 ÷ 2 = 19. (b) At rest the rate is 8 × 2 = 16 breaths per minute, so the step-ups increased the rate from 16 to 38 breaths per minute. Physical activity also makes each breath deeper, so more air is moved per breath. (c) One count can be affected by chance, for example by talking or by losing count. Repeating the measurements and working out a mean makes the results more reliable.

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15 · Part 1 Core Gas exchange in humans [6]

Mark scheme

(a) **[2]**

- 30 × 2.40 (1)
- 72 dm^3 per minute (1)
- the unit is needed for the second mark

(b) **[2]**

- volumes per minute 6.0 (dm^3) for sitting and 21.6 (dm^3) for walking (1)
- 260% (1)

ALLOW (21.6 − 6.0) ÷ 6.0 × 100 for the first mark

(c) **[2]**

- the volume of air per breath / depth of breathing also increases, from 0.50 to 2.40 dm^3 (1)
- if only the rate increased, the volume per minute would be 30 × 0.50 = 15 dm^3 (not 72 dm^3) / the volume per breath increases 4.8 times while the rate increases only 2.5 times (1)

ECF from (a)

Worked solution

(a) Volume per minute = breathing rate × volume per breath = 30 × 2.40 = 72 dm^3 per minute. Check: 72 ÷ 30 = 2.4. (b) Sitting: 12 × 0.50 = 6.0 dm^3 per minute. Walking: 18 × 1.20 = 21.6 dm^3 per minute. Increase = 21.6 − 6.0 = 15.6 dm^3. Percentage increase = 15.6 ÷ 6.0 × 100 = 260%. Check: 6.0 × 3.6 = 21.6, and 3.6 is 260% more than 1. (c) Running increases both the rate and the depth of breathing. If only the rate had changed, keeping 0.50 dm^3 per breath, she would breathe 30 × 0.50 = 15 dm^3 per minute, far less than 72 dm^3. Most of the extra air comes from deeper breaths: the volume per breath goes up 4.8 times.

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16 · Part 1 Extended Gas exchange in humans [6]

Mark scheme

(a) **[4]**

- (external) intercostal muscles contract, moving the ribs upwards and outwards (1)
- diaphragm contracts and flattens / moves down (1)
- volume of the thorax increases (1)
- pressure in the thorax / lungs decreases (below atmospheric pressure), so air is pushed into the lungs (1)

REJECT lungs expand and suck air in (with no reference to pressure)

(b) **[1]**

- ribs move downwards and inwards (1)

(c) **[1]**

- diaphragm relaxes and moves up / becomes dome-shaped (1)

Worked solution

(a) To breathe in, the external intercostal muscles between the ribs contract and pull the ribcage upwards and outwards. At the same time the diaphragm contracts and flattens, moving downwards. Both movements increase the volume of the thorax. A larger volume means a lower pressure, so the pressure inside the lungs falls below the pressure of the air outside and air moves in through the trachea. (b) The internal intercostal muscles pull the ribs downwards and inwards, which reduces the volume of the thorax and forces air out quickly. (c) When breathing out, the diaphragm relaxes and returns to its dome shape, moving upwards and reducing the volume of the thorax.

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17 · Part 1 Extended Gas exchange in humans [5]

Mark scheme

(a) **[2]**

- volume (of the thorax) increases (1)
- pressure (in the thorax) decreases (1)

(b) **[3]**

- air enters the thorax through the hole (when the chest expands) (1)
- so the pressure around the left lung does not fall / stays the same as the air outside (1)
- so there is no pressure difference to push air into the left lung (1)

ALLOW the left lung collapses

Worked solution

(a) When a healthy person breathes in, the ribs move up and out and the diaphragm flattens, so the volume of the thorax increases and the pressure inside it decreases. Air outside is now at a higher pressure, so it moves into the lungs. (b) This only works because the thorax is airtight. With a hole in the chest wall, air rushes in through the hole as the thorax gets bigger, so the pressure around the left lung stays the same as the pressure outside. With no lower pressure around it, nothing makes air move down the trachea into the left lung, so it does not fill (it collapses). The right lung is in a separate airtight space, so it can still work.

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18 · Part 1 Extended Gas exchange in humans [6]

Mark scheme

(a) **[2]**

- oxygen diffuses from the air in the alveoli into the blood (1)
- oxygen is carried away by the blood / used in (aerobic) respiration by cells, so the concentration in the blood stays lower (1)

(b) **[2]**

- carbon dioxide is produced by (aerobic) respiration in cells / tissues and carried to the lungs in the blood (1)
- carbon dioxide diffuses from the blood into the air in the alveoli (1)

(c) **[2]**

- the air in the trachea and bronchi has the same composition as inspired air / has not taken part in gas exchange (1)
- it mixes with alveolar air, raising the oxygen percentage and lowering the carbon dioxide percentage (1)

ALLOW alveolar air is diluted by inspired air

Worked solution

(a) The blood arriving at the alveoli has a low oxygen concentration because cells have used oxygen for aerobic respiration. Oxygen diffuses from the air in the alveoli into the blood, down its concentration gradient, so the air left in the alveoli has less oxygen than the air breathed in. (b) Respiration in cells all over the body produces carbon dioxide, which the blood carries to the lungs. The concentration of carbon dioxide in the blood is higher than in the alveoli, so it diffuses into the alveoli. (c) The last air breathed in fills the trachea and bronchi but never reaches the alveoli, so it still has the composition of inspired air (21% oxygen, 0.04% carbon dioxide). When the person breathes out, this air mixes with the alveolar air on the way out, so expired air has a little more oxygen (16% not 14%) and a little less carbon dioxide (4% not 5%) than the air in the alveoli.

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19 · Part 1 Extended Gas exchange in humans [5]

Mark scheme

(a) **[2]**

- muscles contract more and need more energy, so the rate of (aerobic) respiration in muscle cells increases (1)
- more carbon dioxide is produced and passes into the blood (1)

(b) **[1]**

- brain (1)

(c) **[2]**

- the rate and depth of breathing increase (1)
- less carbon dioxide diffuses out of the blood (into the alveoli), so the carbon dioxide concentration in the blood increases (and is detected by the brain) (1)

ALLOW the concentration gradient of carbon dioxide between blood and alveoli is reduced

Worked solution

(a) During the race the leg muscles contract much more, so they need more energy. Their cells respire faster, and every extra glucose molecule respired produces extra carbon dioxide, which diffuses into the blood. (b) The brain detects the increased carbon dioxide concentration in the blood and causes an increased rate and greater depth of breathing. (c) Air with 5% carbon dioxide has about as much carbon dioxide as the air in the alveoli, so the concentration gradient between the blood and the alveoli is much smaller. Less carbon dioxide diffuses out of the blood, so its concentration in the blood rises even though the person is not exercising. The brain detects this, so breathing becomes faster and deeper.

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20 · Part 1 Extended Gas exchange in humans [6]

Mark scheme

(a) **[2]**

- goblet cells make / secrete mucus (1)
- mucus traps pathogens / bacteria / dust particles (1)

(b) **[2]**

- cilia beat / waft / move (1)
- moving mucus (with the trapped pathogens and particles) up to the throat (where it is swallowed) (1)

REJECT cilia trap / kill pathogens

(c) **[2]**

- mucus (with trapped pathogens) is not moved out of the airways / builds up in the bronchi and lungs (1)
- pathogens stay in the breathing system and can multiply / reach the alveoli, causing infection (1)

Worked solution

(a) Goblet cells in the lining of the trachea and bronchi make mucus. The sticky mucus traps pathogens, such as bacteria, and particles, such as dust, in the air breathed in. (b) Ciliated cells have tiny hair-like cilia on their surface. The cilia beat together and sweep the mucus, with whatever it has trapped, up the trachea to the throat, where it is swallowed. (c) If the cilia cannot move, the mucus is not carried away. It stays in the bronchi and lungs with the pathogens trapped in it. These pathogens have time to multiply and cause infections, so people with this condition get many more lung infections.

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